If the distance traveled by one wave is taken as one wavelength (λ), and the time taken for this propagation is one time period (T), then

$$\text{Speed (v)} = \frac{\text{One wavelength } (\lambda)}{\text{One time period (T)}} \quad \text{(or)} \quad v = \frac{\lambda}{T}$$

As, $T = \frac{1}{n}$, the speed (v) of sound is also written as, $v = n\lambda$.

The speed of sound remains almost the same for all frequencies in a given medium under the same physical conditions.

Problem 1

A sound wave has a frequency of 2 kHz and wavelength of 15 cm. How much time will it take to travel 1.5 km?

Solution:

Speed, $v = n\lambda$

Here, $n = 2\text{ kHz} = 2000\text{Hz}$

$\lambda = 15\text{ cm} = 0.15\text{ m}$

$v = 0.15 \times 2000 = 300\text{ m s}^{-1}$

$$\text{Time (t)} = \frac{\text{Distance (d)}}{\text{Velocity (v)}}$$

$$t = \frac{1500}{300} = 5\text{ s}$$

The sound will take 5 s to travel a distance of 1.5 km.

Problem 2

What is the wavelength of a sound wave in air at 20° C with a frequency of 22 MHz?

Solution:

$\lambda = v/n$

Here, $v = 344\text{ m s}^{-1}$.

$n = 22\text{ MHz} = 22 \times 10^6\text{ Hz}$

$\lambda = 344/22 \times 10^6 = 15.64 \times 10^{-6}\text{ m} = 15.64\ \mu\text{m}$.

8.5.1 Speed of sound in different media

Sound propagates through a medium at a finite speed. The sound of thunder is heard a little later than the flash of light is seen. So, we can make out that sound travels with a speed which is much less than the speed of light. The speed of sound depends on the properties of the medium through which it travels.

The speed of sound is less in gaseous medium compared to solid medium. In any medium the speed of sound increases if we increase the temperature of the medium. For example the speed of sound in air is 330 m s^{-1} at 0 °C and 340 m s^{-1} at 25 °C. The speed of sound at a particular temperature in various media is listed in Table 8.1.

Table 8.1 Speed of sound in different media at 25° C.

State	Medium	Speed in m s^{-1}
Solids	Aluminum	6420
	Nickel	6040
	Steel	5960
	Iron	5950
	Brass	4700
	Glass	3980
Liquids	Water (Sea)	1531
	Water (distilled)	1498
	Ethanol	1207
	Methanol	1103
Gases	Hydrogen	1284
	Helium	965
	Air	340
	Oxygen	316
	Sulphur dioxide	213

More to Know

Sonic boom: When the speed of any object exceeds the speed of sound in air (330 m s^{-1}) it is said to be travelling at supersonic speed. Bullets, jet, aircrafts etc., can travel at supersonic speeds. When an object travels at a speed higher than that of sound in air, it produces shock waves. These shock waves carry a large amount of energy. The air pressure variations associated with this type of shock waves produce a very sharp and loud sound called the 'sonic boom'. The shock waves produced by an aircraft have energy to shatter glass and even damage buildings.

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